EiGENCOMM
More than connection

SDK DEVELOPMENT

AN0026- EC NB-IoT NVRAM Introduction_V1.0

EigenCOMM Wireless Microcontroller

Document Description

This document introduces how does eigencomm nvram work and mechanism used to make calibration and other reliable data fail safe.

Contents

1. Related Information

This document introduce how does nvram work and how to protect data when abnormal condition triggers such as suddenly power off.

This document is only a general description for SDK user and customer, not a detail design document.

1.1 Related Software Resources

EC NB-IoT SDK and Eigencomm EC NB-IoT AT command manual.

2. General Description

In IOT and cellular device, some data is very import, which should be stored reliably and fail safe, for example RF calibration data and IMEI/SN. A safety storage module called nvram is designed in EC NB-IoT SDK. This storage mechanism can ensure data is safe, e.g. in case of suddenly power off or bit-flip situation.

2.1 Purpose

Save calibration or factory used data(IMEI/SN) in flash.

Protect data when power-off suddenly or bit-flip.

NVRAM is not designed for normal data saving, normal user data should be saved in file system.

2.2 NVRAM Layout

There are three regions related to nvram in flash. Shown as figure 1.

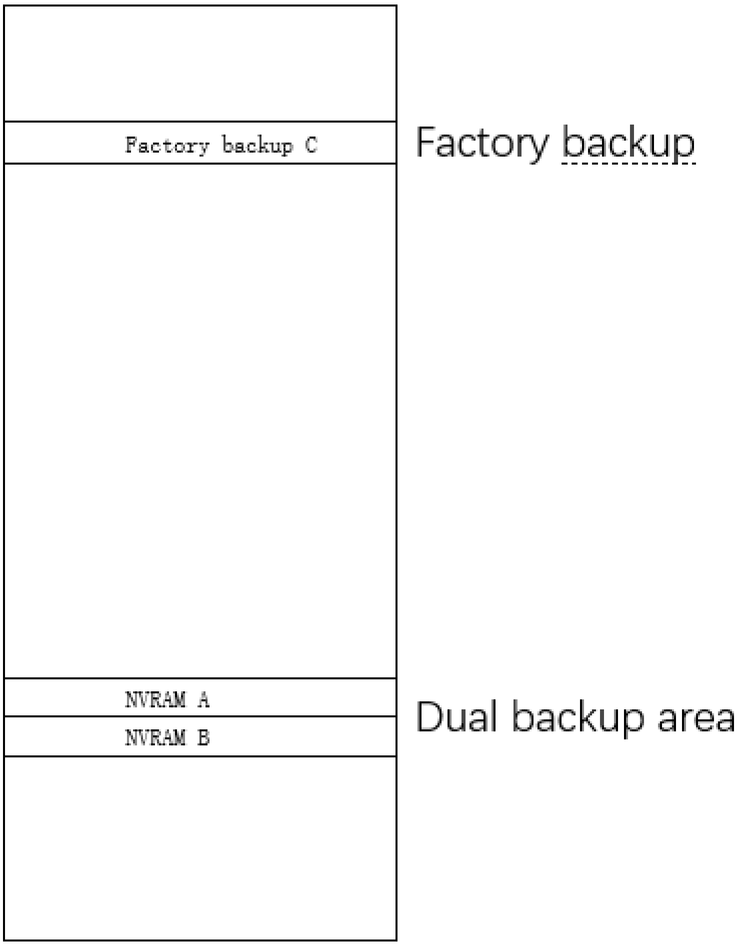

figure 1. NVRAM flash layout

Factory backup region: after production line operation, original RF calibration data and other important data such as IMEI/SN is backup in this region. This region's location is far away from another 2 regions (NVRAMA&B),

to prevent physical damage in consecutive flash regions. The backup region is designed to restore to original factory setting when both NVRAM A and B are broken.

NVRAM A/B region: run time used dual record regions, working as ping-pong buffer. During write operation, always save to “old” one, after writing “old” becomes to “new”. And during read operation, always get data from “new” one. This mechanism is mainly used for power lost protection during normal operation.

Each region has 10 items, and each item has 4K bytes. In current design items 1-8 are used to save RF calibration data. Item 9 is used to save IMEI and SN. Item 10 is reserved. Shown in figure 2.

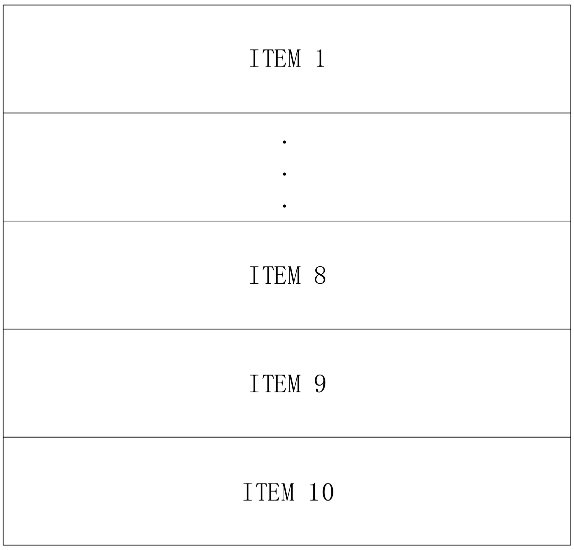

figure 2. Item structure

There is a header in each item. This header contains some data structure to mark record counter(old/new), data size, raw data checksum and etc. Some operation along with the header will ensure reliable data fail safe.

2.3 Working detail

For RF calibration data saving:

Step 1: write default calibration data in NVRAM A using flash download tool. This action makes UE startup and further calibration correctly.

Step 2: when calibration is done, the NPI tool will generate new calibration data, and save this data in item 1-8 in both NVRAM A/B ping-pong saving regions.

Step3: trigger a function to save the calibration data to factory backup region.

For IMEI and SN:

Step1: Send a AT command to set IMEI or SN for UE.

Step2: IMEI/SN will be saved to item9 in both NVRAM A/B ping-pong saving regions.

Step3: save IMEI/SN to factory backup region.

How to recover

During read operation, read new item and validate the data, if data is broken try to read old item. If read one item in region B, and it is broken, then read the same item in A, if A is valid, restore valid data from A to B. If both A and B are broken, it will trigger recover mechanism from factory backup region.

For example, if the IMEI number in item9 of region B is broken. First try to read item9 in region A. If item9 in region A is valid, return the IMEI and restore valid data from A to B (item9). If item9 in region A is also broken, it will recovery IMEI data from item9 in factory backup region and return valid IMEI.

In worst case when A, B, C are all broken, there are some default values saving in code segment that can do recovery to ensure chip can boot up.

3. Test Guild

3.1 Crack Test

This case can crack nvram data directly. when do this test need add some code to compile.

1 enable test code in PLAT\middleware\eigencomm\at\atcust\src\atec_cust_dev.c, find “#if 0//only used for internal test”,chang it to “#if 1//only used for internal test”, shown the picture as below

```
#if 1//only used for internal test
void hexdump(unsigned char *buf, const int num)
{
    int i;
    for(i = 0; i < num; i++)
    {
```

2 uncomment code in PLAT\middleware\eigencomm\at\atcust\src\atec_cust_cmd_table.c, find the lines contain ”ECUNITTEST”, and then uncomment this lines, there are two location need to change, shown the picture as below

```
//AT+ECUNITTEST
const static AtValueAttr attrECUNITTEST[] = { AT_PARAM_ATTR_DEF(AT_DEC_VAL,  AT_MUST_VAL)};
```

```
    AT_CMD_PRE_DEFINE("+ECRFSTAT",      pdevECRFSTAT,    NULL,            AT_EXT_ACT_CMD,     AT_DEFAULT_TIMEOUT_SEC),
    AT_CMD_PRE_DEFINE("+ECUNITTEST",  pdevECUNITTEST, attrECUNITTEST, AT_EXT_PARAM_CMD,   AT_DEFAULT_TIMEOUT_SEC),
    //AT CMD PRE DEFINE("+DUMPDATA",          dbgDUMPDATA,         NULL,                  ATI DUMPDATA,         PNU
```

3 recompile the project, burn it to the target board, you can enable AT+ECUNITTEST command. in the test sample code,

AT+ECUNITTEST=3 can write 4000 bytes with data 0x5A to nvitem 10 in endless loop.

AT+ECUNITTEST=4 can read nvitem 10 is 4000 bytes 0x5A or not

AT+ECUNITTEST=5 can crack data in nvitem 10.

4 we can do some the crack to check recovery, for example. First send AT+ECUNITTEST=3 make nvitem 10 filled 0x5A, then reset the target board, send AT+ECUNITTEST=5 to crack data in nvitem 10. Reset again, send AT+ECUNITTEST=4 to check data is recovery or not.

3.2 Power-off Test

When do the change and enable ecunittest command. We can do the power-off test as below step

1 power on, and send AT+ECUNITTEST=3 to trigger write loop

2 Power-off, then power on send AT+ECUNITTEST=4 to check data is OK or not. Repeat again and again.

If donnt want to do some change, can test as below:

1 use AT+ECCGSN="imei",222222222222222, to set default imei, this command will save imei in nvitem9

2 send AT+ECSAVEFAC="other", to save imei to factory area.

3 now send AT+ECCGSN="imei",111122222222222,to set imei with the different value in endless loop.

4 power-off and the power on the target board, and use AT+CGSN=1,to check imei is correct or recovery.

4. Version

Version	Date	Comments
V1.0	2020-02-11	Draft version

5. About US

Shanghai EigenCOMM Technology Co.,Ltd is established in Feb 2017 in Zhangjiang Hi-Tech park, Shanghai China (www.eigencomm.com) . EigenCOMM focuses on cellular based IoT chipset and solution. With accumulation of experience over the years in algorithm, L1/L2/L3 protocol, SoC, RF , platform & application software as well as the communication architecture and system, team makes its way to IoT industry.

The ultra-low power NB-IoT chipset EC616(S) has already enter mass production stage. The 4G Cat.1 chipset EC618 is in engineering samples stage, 5G is also in the R&D stage.

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